Chapter 12

Integral Calculus

Why Should We Study Integral Calculus?

Integral calculus helps us calculate:

- Area under curves
- Total distance travelled
- Volume and surface area
- Population growth
- Profit and cost accumulation
- Applications in Physics and Engineering

Integration is the reverse process of differentiation.

12.1 Introduction

Integration is one of the fundamental concepts of calculus. It is used to determine quantities that accumulate continuously. If differentiation gives the rate of change, then integration gives the total quantity.

12.2 Indefinite Integrals

Definition of Indefinite Integral

If:

$$\frac{d}{dx}[F(x)] = f(x)$$

then:

$$\int f(x)\,dx = F(x) + C$$

where C is the constant of integration.

12.3 Basic Integration Formulas

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x} \, dx = \ln|x| + C$$

$$\int e^x \, dx = e^x + C$$

$$\int \sin x \, dx = -\cos x + C$$

$$\int \cos x \, dx = \sin x + C$$

Solved Example

Evaluate:

$$\int (3x^2 + 2x + 1) \, dx$$

Solution
Integrate term by term:

$$= 3\int x^2 dx + 2\int x dx + \int 1 dx$$

$$= 3\left(\frac{x^3}{3}\right) + 2\left(\frac{x^2}{2}\right) + x + C$$

$$= x^3 + x^2 + x + C$$

$$\boxed{\int (3x^2 + 2x + 1) \, dx = x^3 + x^2 + x + C}$$

12.4 Definite Integrals

Definition of Definite Integral

A definite integral has upper and lower limits.

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

Solved Example

Evaluate:

$$\int_0^2 x^2\,dx$$

Solution

$$\int x^2 dx = \frac{x^3}{3}$$

Applying limits:

$$= \left[\frac{x^3}{3}\right]_0^2$$

$$= \frac{2^3}{3} - \frac{0^3}{3}$$

$$= \frac{8}{3}$$

$$\boxed{\int_0^2 x^2 dx = \frac{8}{3}}$$

12.5 Geometric Interpretation of Integration

Integration has a very important geometric meaning.
It represents the **area under a curve**.

Suppose:

$$y = f(x)$$

Then:

$$\int_a^b f(x)\,dx$$

gives the area bounded by:

- the curve $y = f(x)$
- the x-axis
- the lines $x = a$ and $x = b$

Simple Real-Life Idea

Imagine rainfall collected over time.

- The rate of rainfall changes continuously.
- Integration adds all small quantities together.
- Thus, integration gives total accumulated rainfall.

Similarly, integration calculates total area, distance, volume and accumulation.

12.6 Area Under a Curve

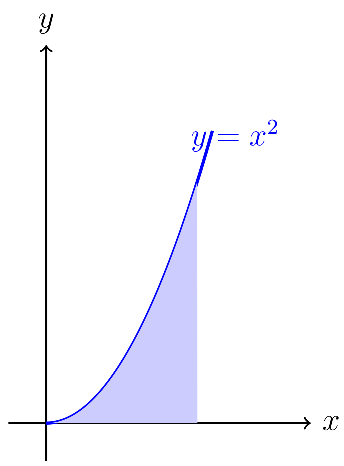
Consider the curve:

$$y = x^2$$

between:

$$x = 0 \quad \text{and} \quad x = 2$$

The shaded region represents the area under the curve.



Area Under the Curve

12.7 Understanding the Graph

The shaded region represents:

$$\int_0^2 x^2\,dx$$

Integration adds infinitely many small rectangles under the curve to calculate total area.

Mathematically:

$$\text{Area} = \int_0^2 x^2 dx$$

Solved Example

Find the area under:

$$y = x^2$$

from:

$$x = 0 \quad \text{to} \quad x = 2$$

Solution

$$\text{Area} = \int_0^2 x^2 dx$$

Integrating:

$$= \left[\frac{x^3}{3} \right]_0^2$$

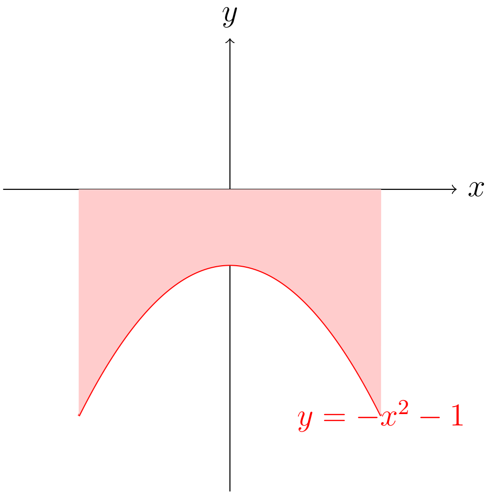
$$= \frac{8}{3} - 0$$

$$= \frac{8}{3}$$

$$\boxed{\text{Area} = \frac{8}{3} \text{ square units}}$$

12.8 Positive and Negative Areas

- If the curve lies above the x-axis, area is positive.
- If the curve lies below the x-axis, area becomes negative.



Negative Area Below the x-axis

12.9 Integration as Accumulation

Integration also represents accumulation of quantities.

Examples:

- Total distance from velocity
- Total profit from profit rate
- Total population growth
- Total water flow

Example

Suppose velocity is:

$$v(t) = 2t$$

Find distance travelled from:

$$t = 0 \quad \text{to} \quad t = 3$$

Solution

Distance:

$$= \int_0^3 2t\,dt$$

$$= \left[t^2\right]_0^3$$

$$= 9$$

$$\boxed{\text{Distance} = 9}$$

12.10 Relation Between Differentiation and Integration

Differentiation and integration are inverse processes.

Differentiation	Integration
Finds rate of change	Finds total accumulation
Slope of curve	Area under curve
Derivative	Antiderivative
$\mathrm{d}_{\overline{dx(x^2)=2x}}$	$\int 2x\,dx = x^2 + C$

Beginner Summary

- Integration represents area under a curve.
- Definite integrals give numerical values.
- Integration adds infinitely small quantities.
- Positive area lies above the x-axis.
- Negative area lies below the x-axis.
- Integration is the reverse process of differentiation.

12.11 Practice Questions

1. Find:

$$\int_0^2 x dx$$

2. Find area under:

$$y = x^2$$

from:

$$x = 1$$

to:

$$x = 3$$

3. Explain geometric meaning of:

$$\int_a^b f(x)\,dx$$

4. What happens if the curve lies below the x-axis?
5. Explain relation between differentiation and integration.

12.12 Practice Questions

1. Find slope of:
$$y = x^2$$
at:
$$x = 2$$
2. Determine where:
$$y = x^3$$
is increasing or decreasing.
3. Find stationary point of:
$$y = x^2 - 4x$$
4. Explain geometrically what:
$$\frac{dy}{dx} = 0$$
means.
5. Draw graph of:
$$y = x^2$$
and mark tangent at:
$$x = 1$$

12.13 Integration by Substitution

This method is used when the integral contains a composite function.

Substitution Method

If:
$$u = g(x)$$
then:
$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

Solved Example

Evaluate:
$$\int 2x(x^2+1)^3 dx$$

Solution
Let:
$$u = x^2 + 1$$

Then:

$$du = 2xdx$$

Therefore:

$$\int 2x(x^2+1)^3dx = \int u^3du$$

$$= \frac{u^4}{4} + C$$

Substitute back:

$$= \frac{(x^2+1)^4}{4} + C$$

$$\boxed{\int 2x(x^2+1)^3dx = \frac{(x^2+1)^4}{4} + C}$$

12.14 Integration by Parts

This method is used when the integral is a product of two functions.

Integration by Parts Formula

$$\int u\,dv = uv - \int v\,du$$

Solved Example

Evaluate:

$$\int xe^x dx$$

Solution
Let:

$$u = x \quad \Rightarrow \quad du = dx$$

$$dv = e^x dx \quad \Rightarrow \quad v = e^x$$

Using formula:

$$\int u\,dv = uv - \int v\,du$$

$$= xe^x - \int e^x dx$$

$$= xe^x - e^x + C$$

$$= e^x(x-1) + C$$

$$\boxed{\int xe^x dx = e^x(x-1) + C}$$

12.15 Area Under Simple Curves

The area under a curve between two limits is calculated using definite integration.

$$\text{Area} = \int_a^b f(x)\,dx$$

Solved Example

Find the area under:

$$y = x^2$$

from:

$$x = 0 \quad \text{to} \quad x = 2$$

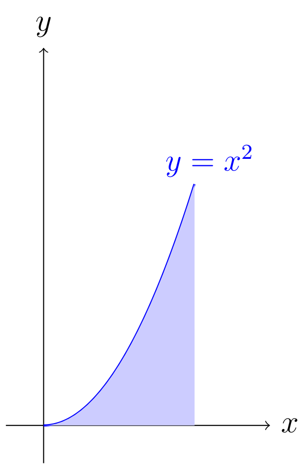
Solution

$$\text{Area} = \int_0^2 x^2 dx$$

$$= \left[\frac{x^3}{3}\right]_0^2$$

$$= \frac{8}{3}$$

$$\boxed{\text{Area} = \frac{8}{3} \text{ square units}}$$



Area Under Curve

Important Notes

- Integration is the reverse process of differentiation.
- Indefinite integrals contain an arbitrary constant C.
- Definite integrals provide numerical values.
- Substitution simplifies complicated integrals.
- Integration by parts is useful for products of functions.
- Definite integration is used to calculate areas and accumulated quantities.

12.16 Solved Questions

Question 1

Evaluate:

$$\int (3x^2 + 4x + 5)\, dx$$

Solution
Integrate term by term:

$$= 3\int x^2 dx + 4\int x dx + \int 5 dx$$

$$= 3\left(\frac{x^3}{3}\right) + 4\left(\frac{x^2}{2}\right) + 5x + C$$

$$= x^3 + 2x^2 + 5x + C$$

$$\boxed{\int (3x^2 + 4x + 5)\, dx = x^3 + 2x^2 + 5x + C}$$

Question 2

Evaluate:

$$\int \left(2x^3 - \frac{1}{x}\right) dx$$

Solution

$$= 2\int x^3 dx - \int \frac{1}{x} dx$$

$$= 2\left(\frac{x^4}{4}\right) - \ln|x| + C$$

$$= \frac{x^4}{2} - \ln|x| + C$$

$$\boxed{\int \left(2x^3 - \frac{1}{x}\right) dx = \frac{x^4}{2} - \ln|x| + C}$$

Question 3

Evaluate:

$$\int_0^2 x^2 dx$$

Solution

$$\int x^2 dx = \frac{x^3}{3}$$

Applying limits:

$$= \left[\frac{x^3}{3}\right]_0^2$$

$$= \frac{8}{3} - 0$$

$$= \frac{8}{3}$$

$$\boxed{\int_0^2 x^2 dx = \frac{8}{3}}$$

Question 4

Evaluate:

$$\int_1^3 (2x + 1)\, dx$$

Solution

$$= \left[x^2 + x\right]_1^3$$

$$= (9 + 3) - (1 + 1)$$

$$= 12 - 2$$

$$= 10$$

$$\boxed{\int_1^3 (2x + 1)\, dx = 10}$$

Question 5

Evaluate using substitution:

$$\int 2x(x^2+1)^5 dx$$

Solution
Let:

$$u = x^2 + 1$$

Then:

$$du = 2x dx$$

Therefore:

$$= \int u^5 du$$

$$= \frac{u^6}{6} + C$$

Substitute back:

$$= \frac{(x^2+1)^6}{6} + C$$

$$\boxed{\int 2x(x^2+1)^5 dx = \frac{(x^2+1)^6}{6} + C}$$

Question 6

Evaluate using substitution:

$$\int \cos x \sin x \, dx$$

Solution
Let:

$$u = \sin x$$

Then:

$$du = \cos x \, dx$$

Thus:

$$= \int u \, du$$

$$= \frac{u^2}{2} + C$$

Substitute back:

$$= \frac{\sin^2 x}{2} + C$$

$$\boxed{\int \cos x \sin x \, dx = \frac{\sin^2 x}{2} + C}$$

Question 7

Evaluate using integration by parts:

$$\int xe^x dx$$

Solution
Let:

$$u = x \Rightarrow du = dx$$

$$dv = e^x dx \Rightarrow v = e^x$$

Using formula:

$$\int u \, dv = uv - \int v \, du$$

$$= xe^x - \int e^x dx$$

$$= xe^x - e^x + C$$

$$= e^x(x - 1) + C$$

$$\boxed{\int xe^x dx = e^x(x - 1) + C}$$

Question 8

Evaluate using integration by parts:

$$\int x \sin x \, dx$$

Solution
Let:

$$u = x \Rightarrow du = dx$$

$$dv = \sin x dx \Rightarrow v = -\cos x$$

Using formula:

$$\int u\,dv = uv - \int v\,du$$

$$= -x\cos x + \int \cos x dx$$

$$= -x\cos x + \sin x + C$$

$$\boxed{\int x\sin x\,dx = -x\cos x + \sin x + C}$$

Question 9

Find the area under the curve:

$$y = x$$

from:

$$x = 0 \quad \text{to} \quad x = 4$$

Solution

$$\text{Area} = \int_0^4 x dx$$

$$= \left[\frac{x^2}{2}\right]_0^4$$

$$= \frac{16}{2}$$

$$= 8$$

$$\boxed{\text{Area} = 8 \text{ square units}}$$

Question 10

Find the area under the curve:

$$y = x^2$$

from:

$$x = 1 \quad \text{to} \quad x = 3$$

Solution

$$\text{Area} = \int_1^3 x^2 dx$$

$$= \left[\frac{x^3}{3}\right]_1^3$$

$$= \frac{27}{3} - \frac{1}{3}$$

$$= \frac{26}{3}$$

$$\boxed{\text{Area} = \frac{26}{3} \text{ square units}}$$

12.17 Practice Questions

1. Evaluate:

$$\int (2x + 5)\, dx$$

2. Evaluate:

$$\int_1^3 x^2 dx$$

3. Evaluate using substitution:

$$\int 3x^2(x^3 + 1)^4 dx$$

4. Evaluate using integration by parts:

$$\int x \sin x\, dx$$

5. Find the area under:

$$y = x^2$$

from:

$$x = 0$$

to:

$$x = 3$$

12.18 Unsolved Exercise

Question 1

Evaluate:

$$\int (4x^3 + 2x + 1)\,dx$$

Question 2

Evaluate:

$$\int \left(x^2 + \frac{1}{x}\right) dx$$

Question 3

Evaluate:

$$\int_0^2 (x + 1)\,dx$$

Question 4

Evaluate:

$$\int_1^3 x^2 dx$$

Question 5

Evaluate using substitution:

$$\int 2x(x^2 + 4)^5 dx$$

Question 6

Evaluate using substitution:

$$\int \sin x \cos x\,dx$$

Question 7

Evaluate using integration by parts:

$$\int xe^x dx$$

Question 8

Evaluate using integration by parts:

$$\int x \cos x \, dx$$

Question 9

Find the area under:

$$y = x^2$$

from:

$$x = 0 \quad \text{to} \quad x = 3$$

Question 10

Find the area under:

$$y = 2x$$

from:

$$x = 1 \quad \text{to} \quad x = 4$$

Answer Key

1. $$x^4 + x^2 + x + C$$

2. $$\frac{x^3}{3} + \ln|x| + C$$

3. $$4$$

4. $$\frac{26}{3}$$

5. $$\frac{(x^2+4)^6}{6} + C$$

6. $$\frac{\sin^2 x}{2} + C$$

7. $$e^x(x-1) + C$$

8. $$x \sin x + \cos x + C$$

9. $$9 \text{ square units}$$

10. $$15 \text{ square units}$$

Practice Reminder

- Practice integration daily.
- Learn formulas carefully.
- Try solving without looking at answers first.
- Draw graphs to understand area concepts better.
- Integration becomes easier with regular practice.